Dense Simultaneous Distributional Chaos and Infinite Entropy in the Full Space of Non-Autonomous Interval Systems

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Abstract

Let $I = [0, 1]$ and let $F(I) = C(I, I)^{\mathbb{N}}$ be the full space of non-autonomous interval systems, equipped with the uniform-in-time metric

$$D(f_{1,\infty}, g_{1,\infty}) = \sup_{n \geq 1} \|f_n - g_n\|_\infty.$$

A recent problem of Balibrea and Rucká asks whether DC1 systems and/or infinite-topological-entropy systems are dense in the full interval non-autonomous dynamical-systems space. We prove a stronger statement: systems which are simultaneously DC1 and of infinite topological entropy are dense in $F(I)$. The same perturbation can be carried out in the subspace of surjective systems. We also prove a prescribed dense-open window version: for every preassigned sequence of dense open sets $W_n \subset I$, the coding intervals may be chosen inside W_n. The proof is constructive and uses small local horseshoes placed in moving windows.

Keywords. Non-autonomous dynamical systems; distributional chaos; DC1 chaos; topological entropy; interval maps; dense perturbations.

Contents

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1 Introduction

A non-autonomous discrete dynamical system on a compact metric space X is a sequence

$$f_{1,\infty} = (f_1, f_2, f_3, \ldots), \qquad f_n \in C(X, X),$$

with orbit

$$x,\ f_1(x),\ f_2(f_1(x)),\ f_3(f_2(f_1(x))), \ldots.$$

The topological entropy of such systems was introduced by Kolyada and Snoha [6]; interval versions and related estimates were further studied by Kolyada, Misiurewicz and Snoha [5]. Distributional chaos, introduced for interval maps by Schweizer and Smítal [7], has been studied in non-autonomous systems in, for example, [4, 8, 9, 1].

Throughout this paper we use the notation

$$F(I) = C(I, I)^{\mathbb{N}}$$

for the full space of all sequences of continuous interval maps, without imposing surjectivity. When the surjective subspace is considered, we write

$$F_s(I) = C_s(I, I)^{\mathbb{N}},$$

where $C_s(I, I)$ denotes the space of continuous surjections $I \to I$. Both spaces are equipped with the uniform-in-time metric

$$D(f_{1,\infty}, g_{1,\infty}) = \sup_{n \geq 1} \sup_{x \in I} |f_n(x) - g_n(x)|.$$

This convention is stated explicitly to avoid ambiguity with related notation in the literature, where surjective and non-surjective interval spaces are sometimes discussed side by side.

Balibrea and Rucká [1], building on problems from [3, 9], prove density of DC1 systems in a pointwise-convergent interval subspace. They also formulate the following open problem for the full space of interval non-autonomous dynamical systems.

Question 1.1 (Balibrea–Rucká, in the notation of this paper). *Are DC1 and/or infinite-topological-entropy systems dense in $F(I)$?*

The purpose of this paper is to answer the question affirmatively in a stronger simultaneous form.

Theorem 1.2 (Main theorem). *Let $I = [0, 1]$, and let $F(I) = C(I, I)^{\mathbb{N}}$ be equipped with D. The set of systems $g_{1,\infty} \in F(I)$ which are both DC1 and have infinite topological entropy is dense in $F(I)$. More precisely, for every $f_{1,\infty} \in F(I)$ and every $\eta > 0$, there exists $g_{1,\infty} \in F(I)$ such that*

$$D(f_{1,\infty}, g_{1,\infty}) < \eta,$$

$g_{1,\infty}$ is DC1, and

$$h(g_{1,\infty}) = \infty.$$

If $f_{1,\infty} \in F_s(I)$, then $g_{1,\infty}$ may be chosen in $F_s(I)$.

The cited problem was posed for the uniform-in-time topology. We treat the full non-surjective sequence space $C(I,I)^{\mathbb{N}}$ and, by a small endpoint-preservation device, obtain the same conclusion inside the surjective subspace $C_s(I,I)^{\mathbb{N}}$.

The construction is flexible enough to respect prescribed admissibility windows. Throughout the paper, "open" and "dense open" subsets of $I=[0,1]$ are understood in the relative topology of I.

Theorem 1.3 (Prescribed dense-open window strengthening)**.** *Let $(W_n)_{n\ge 1}$ be any sequence of dense open subsets of I. In Theorem 1.2, the perturbation $g_{1,\infty}$ may be chosen with coding intervals $B_{n,i}\subset W_n$. Consequently the DC1 scrambled set constructed below has representatives whose coded orbits satisfy*

$$G^{n-1}(x)\in W_n \qquad (n\ge 1),$$

where $G^0=\mathrm{id}$ and $G^n=g_n\circ\cdots\circ g_1$. The same is true for the finite orbit segments used to force infinite topological entropy.

The proof of Theorem 1.3 contains Theorem 1.2 by taking $W_n=I$ for every n. We therefore prove the windowed statement from the beginning. The window version is useful because it shows that the local symbolic structure can be inserted while forcing the coded orbit to visit any prescribed dense open target at each time.

Corollary 1.4. *DC1 systems are dense in $F(I)$, infinite-topological-entropy systems are dense in $F(I)$, and the same holds in the subspace $F_s(I)$ of surjective systems. In particular, DC2, DC3, and positive-topological-entropy systems are dense as well.*

2 Definitions

Throughout the paper,

$$I=[0,1].$$

For a system $f_{1,\infty}$, write

$$F^0=\mathrm{id},\qquad F^n=f_n\circ f_{n-1}\circ\cdots\circ f_1\qquad (n\ge 1).$$

For $N\ge 1$, define the Bowen metric

$$\rho_N(x,y)=\max_{0\le j<N}|F^j(x)-F^j(y)|.$$

A set $E\subset I$ is (N,ε)-separated if $\rho_N(x,y)>\varepsilon$ for all distinct $x,y\in E$. Let $s_N(f_{1,\infty},\varepsilon)$ be the largest cardinality of an (N,ε)-separated subset of I. The non-autonomous topological entropy is

$$h(f_{1,\infty})=\lim_{\varepsilon\downarrow 0}\limsup_{N\to\infty}\frac{1}{N}\log s_N(f_{1,\infty},\varepsilon).$$

For $x,y\in I$ and $t>0$, define the lower and upper distribution functions

$$\psi_{xy}(t)=\liminf_{N\to\infty}\frac{1}{N}\#\{0\le j<N:|F^j(x)-F^j(y)|<t\},$$

and

$$\psi^*_{xy}(t)=\limsup_{N\to\infty}\frac{1}{N}\#\{0\le j<N:|F^j(x)-F^j(y)|<t\}.$$

The pair (x, y) is DC1-scrambled if

$$\psi^*_{xy}(t) = 1 \qquad \text{for every } t > 0,$$

and

$$\psi_{xy}(t_0) = 0 \qquad \text{for some } t_0 > 0.$$

The system is DC1 if it has an uncountable set $S \subset I$ such that every distinct pair in S is DC1-scrambled.

For comparison, recall the usual weaker distributional-chaos notions [2]. A pair (x, y) is DC2-scrambled if

$$\psi^*_{xy}(t) = 1 \qquad \text{for every } t > 0$$

and

$$\psi_{xy}(t_0) < 1 \qquad \text{for some } t_0 > 0.$$

A pair (x, y) is DC3-scrambled if there is a nondegenerate interval $J \subset (0, \infty)$ such that

$$\psi_{xy}(t) < \psi^*_{xy}(t) \qquad (t \in J).$$

A system is DC2, respectively DC3, if it has an uncountable set whose distinct pairs are DC2-scrambled, respectively DC3-scrambled. It is immediate from the definitions that

$$\text{DC1} \Longrightarrow \text{DC2} \Longrightarrow \text{DC3}.$$

3 A selection lemma inside dense open windows

The perturbation will repeatedly need many well-separated points in a small interval whose images under a given continuous map are close. The following elementary lemma is the source of the flexibility.

Lemma 3.1 (Separated points in a set of positive measure)**.** *Let $A \subset [0, 1]$ be Lebesgue measurable and let $\lambda(A) > 0$. For every $m \geq 2$, the set A contains m points whose pairwise distances are strictly larger than*

$$\frac{\lambda(A)}{4m}.$$

Proof. Put $\ell = \lambda(A)$ and $r = \ell/(8m)$. Suppose no such m points exist with pairwise distance larger than $2r = \ell/(4m)$. Then a maximal $2r$-separated subset of A has at most $m - 1$ points. By maximality, A is covered by at most $m - 1$ intervals of radius $2r$, each of length $4r$. Hence

$$\lambda(A) \leq (m - 1)4r < 4mr = \frac{\ell}{2},$$

which is impossible. Thus the desired separated set exists. □

Lemma 3.2 (Windowed small-image selection)**.** *Fix $0 < \alpha < 1$ and $m \geq 2$. There exists $\delta_m > 0$, depending only on α and m, with the following property. Let*

- *$J \subset I$ be a closed interval of length α,*
- *$W \subset I$ be dense and open,*
- *$f \in C(I, I)$, and*

- *$F \subset I$ be finite.*

Then there exist points

$$p_1, \ldots, p_m \in J \cap W \setminus F$$

such that

$$|p_i - p_j| > 4\delta_m \qquad (i \neq j),$$

and

$$\operatorname{diam}\{f(p_1), \ldots, f(p_m)\} < \frac{\alpha}{8}.$$

Moreover, the points may be chosen in the middle subinterval of J of length $\alpha/2$.

Proof. Let J^* be the middle closed subinterval of J of length $\alpha/2$. Partition $[0, 1]$ into

$$K = \left\lceil \frac{16}{\alpha} \right\rceil$$

closed intervals $R_1, \ldots, R_K$ with pairwise disjoint interiors, each of length at most $\alpha/16$. The sets

$$A_j = J^* \cap f^{-1}(R_j), \qquad j = 1, \ldots, K,$$

cover J^*. Hence for some j,

$$\lambda(A_j) \geq \frac{|J^*|}{K} = \frac{\alpha}{2K}.$$

Removing the finite set $F \cup \partial J^*$ does not change the measure. By Lemma 3.1, the set

$$A_j \setminus (F \cup \partial J^*)$$

contains points

$$z_1, \ldots, z_m$$

whose pairwise distances are larger than

$$\frac{1}{4m} \cdot \frac{\alpha}{2K} = \frac{\alpha}{8Km}.$$

Set

$$\delta_m = \frac{\alpha}{64Km}.$$

Then $|z_i - z_j| > 8\delta_m$ for $i \neq j$. Also all $f(z_i)$ lie in the same interval R_j, so

$$\operatorname{diam}\{f(z_1), \ldots, f(z_m)\} \leq \frac{\alpha}{16}.$$

Since the points z_i do not belong to ∂J^*, each z_i lies in the relative interior of J^*. By continuity of f, for each i there is a small relatively open interval U_i around z_i, contained in the relative interior of J^*, such that

$$U_i \cap F = \varnothing,$$

$$\operatorname{dist}(U_i, U_j) > 4\delta_m \qquad (i \neq j),$$

and

$$|f(u) - f(z_i)| < \frac{\alpha}{64} \qquad (u \in U_i).$$

Since W is dense and open, $U_i \cap W \neq \varnothing$. Choose

$$p_i \in U_i \cap W.$$

Then $p_i \notin F$, the points are pairwise more than $4\delta_m$ apart, and

$$\operatorname{diam}\{f(p_1), \ldots, f(p_m)\} \leq \frac{\alpha}{16} + 2 \cdot \frac{\alpha}{64} = \frac{3\alpha}{32} < \frac{\alpha}{8}.$$

This proves the lemma. $\square$

4 The local perturbation scheme

Fix an arbitrary system

$$f_{1,\infty} = (f_1, f_2, \dots) \in F(I),$$

a number $\eta > 0$, and a sequence $(W_n)_{n\geq 1}$ of dense open subsets of I. We construct $g_{1,\infty}$ with $D(f_{1,\infty}, g_{1,\infty}) < \eta$ and with the desired symbolic behavior.

Let

$$\eta_0 = \min\{\eta, 1/2\}, \qquad \alpha = \frac{\eta_0}{20}.$$

For every $m \geq 2$, let δ_m be the constant from Lemma 3.2 for this value of α.

4.1 The schedule of blocks

Let

$$\Lambda = \{C\} \cup \{D_r : r \geq 1\} \cup \{E_m : m \geq 2\}$$

be a countable set of labels. The label C will force long common blocks, D_r will force long disagreement blocks in the r-th binary coordinate, and E_m will force long m-branch entropy blocks.

Choose a sequence

$$\lambda_1, \lambda_2, \lambda_3, \dots \in \Lambda$$

such that every label in Λ occurs infinitely often. Choose positive integers L_k recursively so fast that, with

$$S_k = L_1 + \cdots + L_{k-1} \qquad (S_1 = 0),$$

one has

$$\frac{S_k}{L_k} \longrightarrow 0.$$

For instance, after choosing $L_1, \dots, L_{k-1}$, choose

$$L_k > k^2(S_k + 1).$$

The k-th time block is

$$\mathcal{T}_k = \{S_k + 1, S_k + 2, \dots, S_k + L_k\}.$$

For each time n, let $k(n)$ be the unique block index with $n \in \mathcal{T}_{k(n)}$. Define the number of coding branches at time n by

$$M_n = \begin{cases} 2, & \lambda_{k(n)} = C, \\ 2, & \lambda_{k(n)} = D_r \text{ for some } r, \\ m, & \lambda_{k(n)} = E_m. \end{cases}$$

4.2 Recursive windows and coding intervals

We construct closed intervals

$$J_n \subset I, \qquad |J_n| = \alpha,$$

and, inside $J_n \cap W_n$, closed coding intervals

$$B_{n,1}, \dots, B_{n,M_n}.$$

Start with

$$J_1 = [0, \alpha].$$

Assume J_n is known. Put $m = M_n$.

If we are proving the surjective version and f_n is surjective, choose points $a_n, b_n \in I$ such that

$$f_n(a_n) = 0, \qquad f_n(b_n) = 1,$$

and put

$$F_n = \{a_n, b_n\}.$$

If no surjectivity preservation is required, put $F_n = \varnothing$.

Apply Lemma 3.2 to J_n, W_n, f_n, F_n, and $m = M_n$. We obtain points

$$p_{n,1}, \ldots, p_{n,m} \in J_n \cap W_n \setminus F_n$$

such that

$$|p_{n,i} - p_{n,j}| > 4\delta_m \qquad (i \neq j),$$

and

$$\operatorname{diam}\{f_n(p_{n,1}), \ldots, f_n(p_{n,m})\} < \frac{\alpha}{8}.$$

Because f_n is continuous and W_n is open, choose pairwise disjoint closed intervals

$$D_{n,1}, \ldots, D_{n,m} \subset J_n \cap W_n$$

with

$$p_{n,i} \in \operatorname{int} D_{n,i},$$
$$D_{n,i} \cap F_n = \varnothing,$$
$$\operatorname{dist}(D_{n,i}, D_{n,j}) > 2\delta_m \qquad (i \neq j),$$

and

$$\operatorname{diam} f_n(D_{n,1} \cup \cdots \cup D_{n,m}) < \frac{\alpha}{4}.$$

Now choose smaller nondegenerate closed intervals

$$B_{n,i} \subset \operatorname{int} D_{n,i}$$

so that

$$\operatorname{diam}(B_{n,i}) < \frac{1}{n},$$

and

$$\operatorname{dist}(B_{n,i}, B_{n,j}) > \delta_m \qquad (i \neq j).$$

Let

$$c_{n+1} = f_n(p_{n,1}).$$

Choose a closed interval $J_{n+1} \subset I$ of length α containing c_{n+1}. For example, take the centered interval when possible, and use $[0, \alpha]$ or $[1 - \alpha, 1]$ near the endpoints. Then every point of J_{n+1} is within distance at most α of c_{n+1}.

4.3 Definition of g_n

Outside

$$D_{n,1} \cup \cdots \cup D_{n,M_n}$$

set

$$g_n(x) = f_n(x).$$

Fix i. Write

$$D_{n,i} = [u, v], \qquad B_{n,i} = [r, s], \qquad J_{n+1} = [A, B],$$

with $u < r < s < v$. On $D_{n,i}$, define g_n as the piecewise linear map determined by

$$g_n(u) = f_n(u), \qquad g_n(r) = A,$$

$$g_n(s) = B, \qquad g_n(v) = f_n(v),$$

and by linearity on $[u, r]$, $[r, s]$, and $[s, v]$. Then

$$g_n(B_{n,i}) = J_{n+1}.$$

Since $g_n = f_n$ at the endpoints of every $D_{n,i}$, the pieces glue continuously. Thus $g_n \in C(I, I)$.

Lemma 4.1 (Uniform smallness of the perturbation)**.** *For every n,*

$$\|g_n - f_n\|_\infty < \frac{5\alpha}{4} = \frac{\eta_0}{16}.$$

Consequently

$$D(f_{1,\infty}, g_{1,\infty}) \leq \frac{\eta_0}{16} < \eta.$$

Proof. Outside $\bigcup_i D_{n,i}$, the maps are equal. Let $x \in D_{n,i}$. By construction, all values of f_n on $\bigcup_i D_{n,i}$ are within distance $< \alpha/4$ of

$$c_{n+1} = f_n(p_{n,1}).$$

Also every point of J_{n+1} is within distance at most α of c_{n+1}.

The values used to define g_n on $D_{n,i}$ are the endpoint values $f_n(u), f_n(v)$, which are within $\alpha/4$ of c_{n+1}, and the two endpoint values A, B of J_{n+1}, which are within α of c_{n+1}. Hence every value $g_n(x)$ on $D_{n,i}$, being a convex combination of these values on each linear piece, is within α of c_{n+1}. Therefore

$$|g_n(x) - f_n(x)| < \alpha + \frac{\alpha}{4} = \frac{5\alpha}{4}.$$

Taking the supremum over $x \in I$ gives the first estimate. Taking the supremum over n gives

$$D(f_{1,\infty}, g_{1,\infty}) \leq \frac{5\alpha}{4} = \frac{\eta_0}{16} < \eta.$$

□

Lemma 4.2 (Surjectivity is preserved)**.** *If f_n is surjective and the construction is made with $F_n = \{a_n, b_n\}$ as above, then g_n is surjective. Hence if every f_n is surjective, every g_n can be chosen surjective.*

Proof. The modification intervals $D_{n,i}$ were chosen disjoint from $F_n = \{a_n, b_n\}$. Thus

$$g_n(a_n) = f_n(a_n) = 0, \qquad g_n(b_n) = f_n(b_n) = 1.$$

Since g_n is continuous and attains both endpoint values 0 and 1, the intermediate value theorem gives $g_n(I) = I$. □

5 Symbolic coding

From now on, let

$$G^0 = \mathrm{id}, \qquad G^n = g_n \circ g_{n-1} \circ \cdots \circ g_1 \qquad (n \geq 1).$$

The key property is

$$B_{n,i} \subset J_n \cap W_n, \qquad g_n(B_{n,i}) = J_{n+1}.$$

Lemma 5.1 (Infinite itinerary coding)**.** *For every sequence*

$$\omega = (\omega_1, \omega_2, \ldots), \qquad \omega_n \in \{1, \ldots, M_n\},$$

there exists $x_\omega \in J_1$ such that

$$G^{n-1}(x_\omega) \in B_{n,\omega_n} \subset W_n \qquad (n \geq 1).$$

Moreover, if two itineraries differ at some time n, then no point can realize both itineraries.

Proof. First fix $N \geq 1$. We claim there exists $x \in J_1$ such that

$$G^{n-1}(x) \in B_{n,\omega_n} \qquad (1 \leq n \leq N).$$

If $N = 1$, choose any $y_1 \in B_{1,\omega_1}$. Now assume $N \geq 2$. Choose any $y_N \in B_{N,\omega_N}$. Since

$$g_{N-1}(B_{N-1,\omega_{N-1}}) = J_N$$

and $y_N \in B_{N,\omega_N} \subset J_N$, there exists $y_{N-1} \in B_{N-1,\omega_{N-1}}$ with $g_{N-1}(y_{N-1}) = y_N$. Continuing backward, we obtain $y_1 \in B_{1,\omega_1} \subset J_1$ with the desired finite itinerary.

Thus the compact sets

$$K_N(\omega) = \{x \in J_1 : G^{n-1}(x) \in B_{n,\omega_n} \text{ for } 1 \leq n \leq N\}$$

are nonempty, closed, and nested. They are closed because each B_{n,ω_n} is closed and each G^{n-1} is continuous. Their intersection is nonempty by compactness, and every point in the intersection realizes the infinite itinerary.

If two itineraries differ at time n, then a point realizing both would have $G^{n-1}(x)$ in two distinct intervals $B_{n,i}$, which are disjoint. This is impossible. $\square$

Corollary 5.2 (Finite itinerary coding)**.** *For every $N \geq 1$ and every finite admissible word*

$$(\omega_1, \ldots, \omega_N), \qquad \omega_n \in \{1, \ldots, M_n\},$$

there exists $x \in J_1$ such that

$$G^{n-1}(x) \in B_{n,\omega_n} \qquad (1 \leq n \leq N).$$

Proof. This is exactly the finite-itinerary claim proved in the first paragraph of Lemma 5.1. $\square$

6 DC1 chaos

We now define an uncountable family of itineraries. Let

$$\beta = (\beta_1, \beta_2, \ldots) \in \{0,1\}^{\mathbb{N}}.$$

Define ω^β block by block.

If $n \in \mathcal{T}_k$ and $\lambda_k = C$, set

$$\omega_n^\beta = 1.$$

Thus all itineraries agree on every C-block.

If $n \in \mathcal{T}_k$ and $\lambda_k = D_r$, set

$$\omega_n^\beta = 1 + \beta_r.$$

Thus on a D_r-block the itinerary records the r-th bit of β, constantly across the whole block.

If $n \in \mathcal{T}_k$ and $\lambda_k = E_m$, set

$$\omega_n^\beta = 1.$$

Entropy blocks are ignored by the DC1 coding.

For each β, choose one point $x_\beta = x_{\omega^\beta}$ supplied by Lemma 5.1, and put

$$S = \{x_\beta : \beta \in \{0,1\}^{\mathbb{N}}\}.$$

Proposition 6.1. *The set S is uncountable.*

Proof. It is enough to show that $\beta \mapsto x_\beta$ is injective. Let $\beta \neq \gamma$. Choose $r \geq 1$ such that $\beta_r \neq \gamma_r$. Since the label D_r occurs infinitely often, there is a block $\mathcal{T}_k$ with $\lambda_k = D_r$. On that block, the two itineraries take different values, one equal to 1 and the other equal to 2. By the last assertion of Lemma 5.1, no single point realizes both itineraries. Hence $x_\beta \neq x_\gamma$. □

Proposition 6.2. *Every distinct pair of points in S is DC1-scrambled. Hence $g_{1,\infty}$ is DC1.*

Proof. Take $\beta \neq \gamma$. We prove the two DC1 conditions.

First let $t > 0$. Since $\operatorname{diam}(B_{n,i}) < 1/n$, there exists N_t such that

$$\operatorname{diam}(B_{n,i}) < t$$

for all $n \geq N_t$ and all admissible i. Consider a block $\mathcal{T}_k$ with $\lambda_k = C$ and $S_k + 1 \geq N_t$. There are infinitely many such blocks. For every $n \in \mathcal{T}_k$, both itineraries have value 1, so

$$G^{n-1}(x_\beta),\ G^{n-1}(x_\gamma) \in B_{n,1}.$$

Consequently

$$|G^{n-1}(x_\beta) - G^{n-1}(x_\gamma)| < t$$

for all $n \in \mathcal{T}_k$. Therefore, among the first $S_k + L_k$ iterates, at least L_k are t-close:

$$\frac{1}{S_k + L_k} \#\{0 \leq j < S_k + L_k : |G^j(x_\beta) - G^j(x_\gamma)| < t\} \geq \frac{L_k}{S_k + L_k}.$$

Along these C-blocks,

$$\frac{S_k}{L_k} \to 0,$$

so

$$\frac{L_k}{S_k + L_k} \to 1.$$

Thus

$$\psi^*_{x_\beta x_\gamma}(t) = 1.$$

Since $t > 0$ was arbitrary,

$$\psi^*_{x_\beta x_\gamma}(t) = 1 \qquad \text{for all } t > 0.$$

Now choose $r \geq 1$ with $\beta_r \neq \gamma_r$. Consider blocks $\mathcal{T}_k$ with $\lambda_k = D_r$. On such a block, for every $n \in \mathcal{T}_k$, one orbit point lies in $B_{n,1}$ and the other lies in $B_{n,2}$. Since $M_n = 2$ on D_r-blocks, these intervals are separated by more than δ_2. Hence

$$|G^{n-1}(x_\beta) - G^{n-1}(x_\gamma)| > \delta_2$$

for all $n \in \mathcal{T}_k$. Put

$$t_0 = \delta_2.$$

Then during the entire block $\mathcal{T}_k$, the two points are not t_0-close. Thus

$$\frac{1}{S_k + L_k} \#\{0 \leq j < S_k + L_k : |G^j(x_\beta) - G^j(x_\gamma)| < t_0\} \leq \frac{S_k}{S_k + L_k}.$$

Along the infinitely many D_r-blocks, the right-hand side tends to 0. Therefore

$$\psi_{x_\beta x_\gamma}(t_0) = 0.$$

The pair is DC1-scrambled. □

7 Infinite topological entropy

We next use the E_m-blocks.

Proposition 7.1. *The constructed system has infinite topological entropy:*

$$h(g_{1,\infty}) = \infty.$$

Proof. Fix $m \geq 2$. Since E_m occurs infinitely often in the label sequence, there are infinitely many blocks $\mathcal{T}_k$ with

$$\lambda_k = E_m.$$

Fix such a block. Its time interval is

$$\mathcal{T}_k = \{S_k + 1, \ldots, S_k + L_k\}.$$

During this block, $M_n = m$.

For every word

$$u = (u_{S_k+1}, \ldots, u_{S_k+L_k}) \in \{1, \ldots, m\}^{L_k},$$

choose a full finite itinerary up to time $S_k + L_k$ by prescribing this word on the block and prescribing symbol 1 outside the block. By Corollary 5.2, there is a point $x_u \in J_1$ such that

$$G^{n-1}(x_u) \in B_{n,u_n} \qquad (n \in \mathcal{T}_k),$$

where u_n denotes the block symbol at time n.

If $u \neq v$, then they differ at some $n \in \mathcal{T}_k$. At that time,

$$G^{n-1}(x_u) \in B_{n,u_n}, \qquad G^{n-1}(x_v) \in B_{n,v_n},$$

with $u_n \neq v_n$. The intervals $B_{n,1}, \ldots, B_{n,m}$ are pairwise separated by more than δ_m. Hence

$$|G^{n-1}(x_u) - G^{n-1}(x_v)| > \delta_m.$$

Since $n - 1 < S_k + L_k$, the set

$$E_k = \{x_u : u \in \{1, \ldots, m\}^{L_k}\}$$

is $(S_k + L_k, \delta_m)$-separated. It has cardinality m^{L_k}. Therefore

$$s_{S_k+L_k}(g_{1,\infty}, \delta_m) \geq m^{L_k}.$$

Taking logarithms gives

$$\frac{1}{S_k + L_k} \log s_{S_k+L_k}(g_{1,\infty}, \delta_m) \geq \frac{L_k}{S_k + L_k} \log m.$$

Along the infinitely many E_m-blocks,

$$\frac{S_k}{L_k} \to 0,$$

so

$$\limsup_{N\to\infty} \frac{1}{N} \log s_N(g_{1,\infty}, \delta_m) \geq \log m.$$

Let

$$H(\varepsilon) = \limsup_{N\to\infty} \frac{1}{N} \log s_N(g_{1,\infty}, \varepsilon).$$

Since $s_N(g_{1,\infty}, \varepsilon)$ is nonincreasing as a function of ε, the function $H(\varepsilon)$ is nonincreasing in ε. Hence

$$h(g_{1,\infty}) = \lim_{\varepsilon\downarrow 0} H(\varepsilon) = \sup_{\varepsilon>0} H(\varepsilon).$$

The separation scale δ_m depends on m, but this is harmless because entropy is the supremum over all positive scales. In particular,

$$h(g_{1,\infty}) \geq H(\delta_m) \geq \log m.$$

The integer $m \geq 2$ was arbitrary. Hence

$$h(g_{1,\infty}) = \infty.$$

□

8 Proofs of the main results

Proof of Theorems 1.2 and 1.3. Given $f_{1,\infty}$, $\eta > 0$, and dense open windows (W_n), Sections 4–7 construct a system $g_{1,\infty}$ with coding intervals $B_{n,i} \subset W_n$. Lemma 4.1 gives

$$D(f_{1,\infty}, g_{1,\infty}) < \eta.$$

Lemma 4.2 gives the surjective version. Lemma 5.1 and Corollary 5.2 prove the advertised windowed itinerary statements. Proposition 6.2 proves DC1 chaos, and Proposition 7.1 proves infinite topological entropy. Thus Theorem 1.3 is proved. Taking $W_n = I$ for all n gives Theorem 1.2. ☐

Proof of Corollary 1.4. Theorem 1.2 proves density of systems which are simultaneously DC1 and of infinite topological entropy. Hence the individual classes are dense. Since DC1 implies DC2 and DC2 implies DC3, the corresponding distributional-chaos classes are dense as well. Infinite topological entropy implies positive topological entropy, so positive-topological-entropy systems are dense. ☐

9 Remarks and related directions

Remark 9.1 (Why the full-space problem is easier than it first appears). The full metric D is strong because it requires $\|f_n - g_n\|_\infty$ to be small uniformly in n. However, it does not impose any convergence condition on the tail. This allows one to insert a small local coding window at every time. The selection lemma guarantees that, inside any interval of fixed length α, finitely many well-separated points have images lying in a small range interval. This is enough to modify f_n by less than η while mapping several tiny subintervals onto the next window.

Remark 9.2 (What is not proved here). This note proves density, not residuality, in $F(I)$. It does not show that DC1 systems or infinite-entropy systems form a dense G_δ subset of $F(I)$. The construction also uses non-invertible interval maps in an essential way and is not a homeomorphic construction.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

References

[1] F. Balibrea and L. Rucká, Density of distributional chaos in non-autonomous systems, *Topology Appl.* **380** (2026), 109735. DOI: https://doi.org/10.1016/j.topol.2026.109735.

[2] F. Balibrea, J. Smítal and M. Štefánková, The three versions of distributional chaos, *Chaos Solitons Fractals* **23** (2005), 1581–1583.

[3] F. Balibrea, J. Smítal and M. Štefánková, On generic properties of nonautonomous dynamical systems, *Internat. J. Bifur. Chaos Appl. Sci. Engrg.* **28** (2018), no. 8, 1850102.

[4] J. Dvořáková, Chaos in nonautonomous discrete dynamical systems, *Commun. Nonlinear Sci. Numer. Simul.* **17** (2012), 4649–4652.

[5] S. Kolyada, M. Misiurewicz and Ľ. Snoha, Topological entropy of nonautonomous piecewise monotone dynamical systems on the interval, *Fund. Math.* **160** (1999), 161–181.

[6] S. Kolyada and Ľ. Snoha, Topological entropy of nonautonomous dynamical systems, *Random Comput. Dynam.* **4** (1996), 205–233.

[7] B. Schweizer and J. Smítal, Measures of chaos and a spectral decomposition of dynamical systems on the interval, *Trans. Amer. Math. Soc.* **344** (1994), 737–754.

[8] H. Shao, Y. Shi and H. Zhu, On distributional chaos in non-autonomous discrete systems, *Chaos Solitons Fractals* **107** (2018), 234–243.

[9] M. Štefánková, Inheriting of chaos in uniformly convergent nonautonomous dynamical systems on the interval, *Discrete Contin. Dyn. Syst.* **36** (2016), 3435–3443.